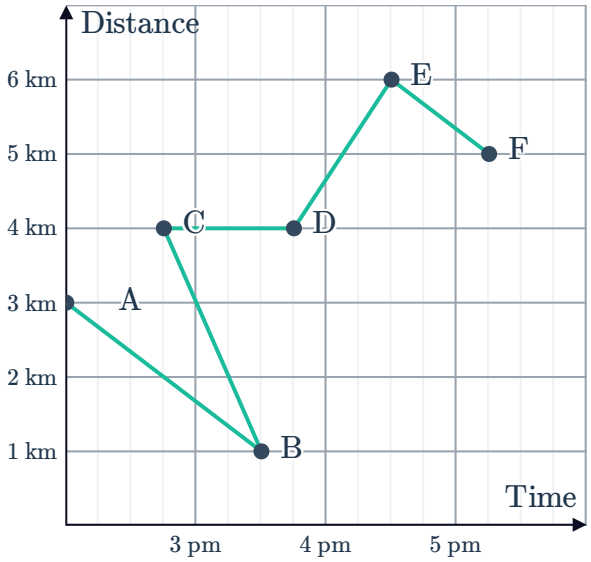
Travel graphs

1 Sketch a graph that shows the height of a ball being thrown vertically into the air. Time should be on the horizontal axis, and height should be on the vertical axis.

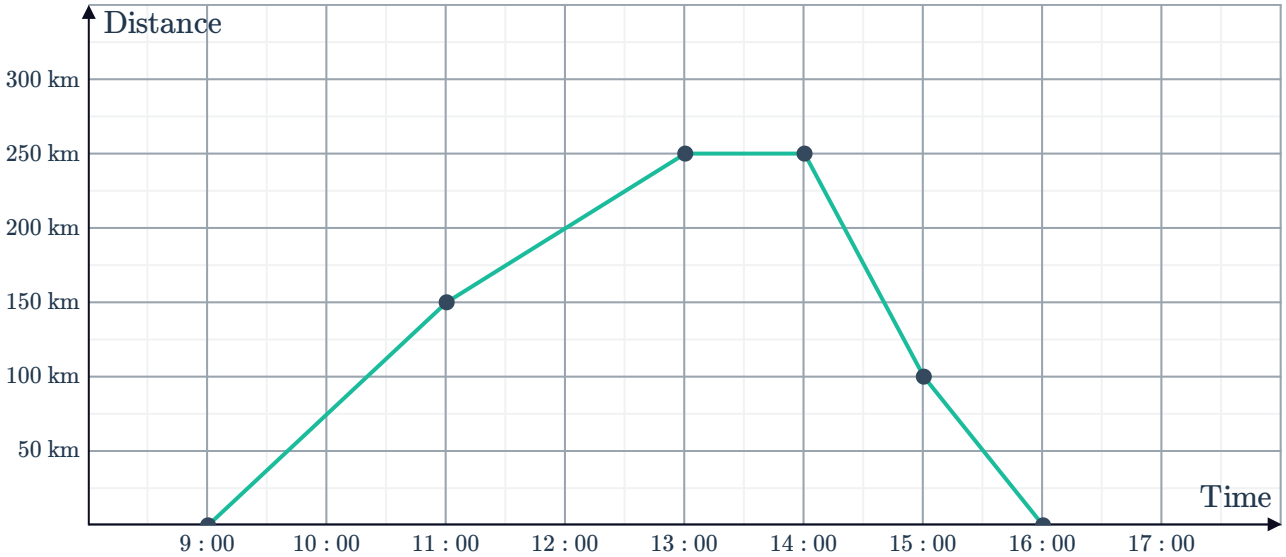
2 A man is driving along a highway in a car at a constant speed. After a period of time, he decreases his speed because of road work. Sketch a graph that represents the relationship between distance travelled and time taken.

3 Tina is cycling to the bus stop, but halfway there she gets a flat tire and has to push her bike the rest of the way. Upon reaching the bus stop, she has a short wait for the bus and then catches the bus to her final destination. Sketch a graph that represents the relationship between distance travelled and time taken.

4 Which line segment of this travel graph is impossible? Explain your answer.



5 Ben travels forwards and backwards along a straight line. The graph shows Ben's distance from his starting point at various times of the day:



a How far did Ben travel by 11 am?

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b What happened to Ben's speed at 11 am?

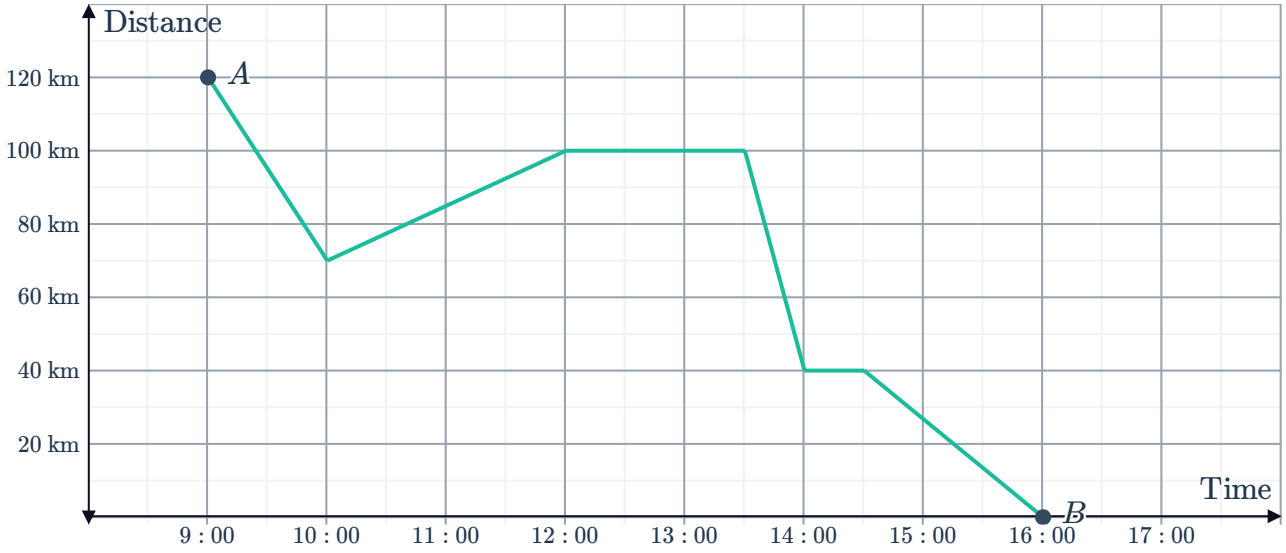
c Evaluate Ben's speed between 11 am and 1 pm.

d What distance did Ben travel between 1 pm and 2 pm?

e What is the furthest distance travelled from the starting point?

f What is the total distance travelled by Ben from 9 am to 4 pm?

6 Ken starts travelling at 9 am from point A to point B. The distance between Ken and point B at various times in his journey is shown on the graph below:



a How far is point B from point A?

b How many hours was Ken's journey from point A to point B?

c State the time period in which Ken travelled at the fastest speed.

d Find Ken's fastest speed in km/h.

e Find the distance in kilometres travelled by Ken between 1:30 pm and 4 pm.

f Find the total distance travelled by Ken from 9 am to 4 pm.

7 The following graph describes Frank's distance from home throughout his journey:

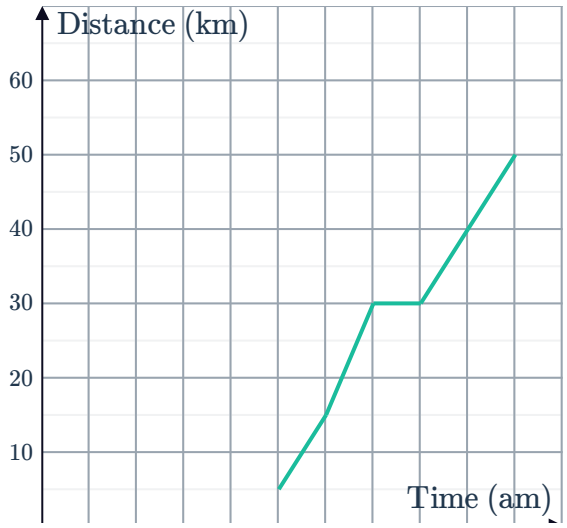
a How far was Frank from home when he started driving at 5 am?

b How far was he from home by 7 am?

c How long did he rest for?

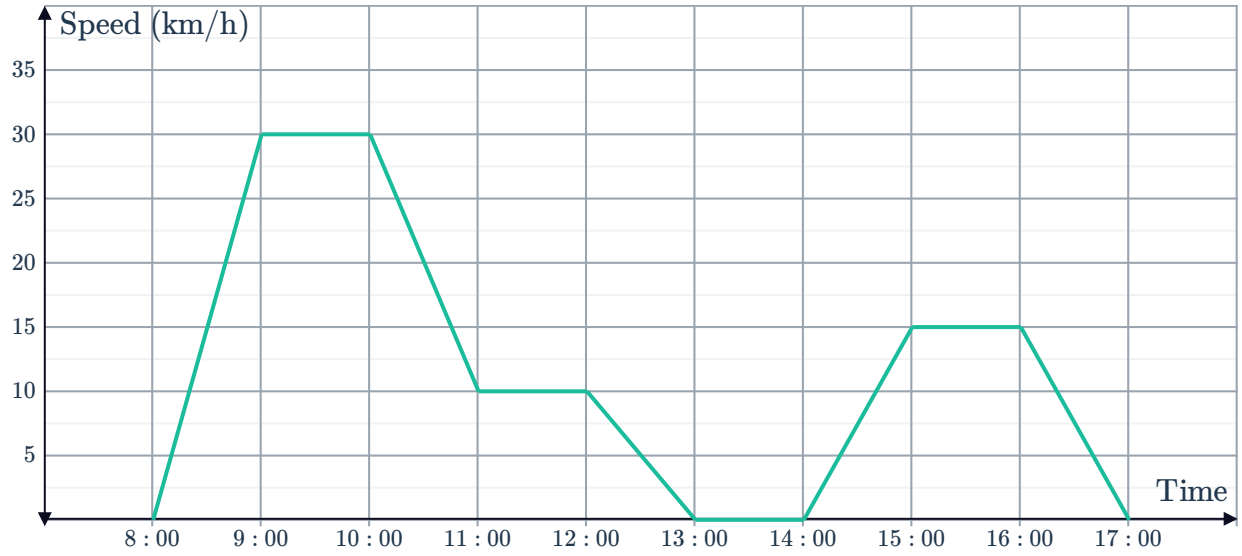
d How far did he travel between 7 am and 9 am?

e Was Frank travelling faster in the first two hours or the last two hours of his trip?



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8 Consider the following travel graph:



a Find the total time taken for the journey.

b Find the total distance covered in the journey.

c Find the average speed during the journey, correct to two decimal places.

9 The graph represents Beth's distance from home over a 9-minute interval:

a When she first started to move, did she travel towards or away from home?

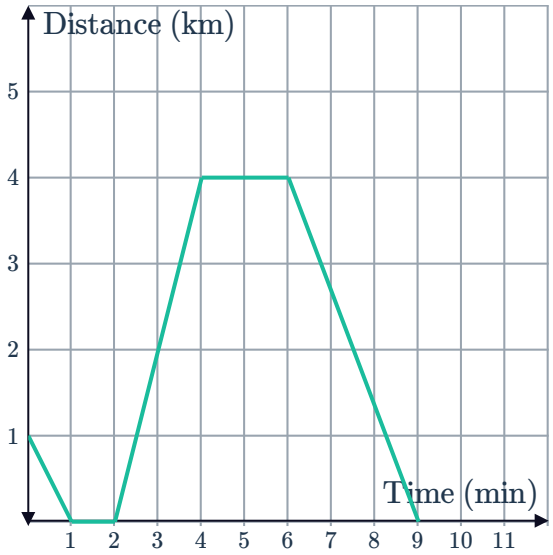
b For how many minutes did she stay home?

c How many minutes into her journey was she when she left home?

d Find Beth's furthest distance from home.

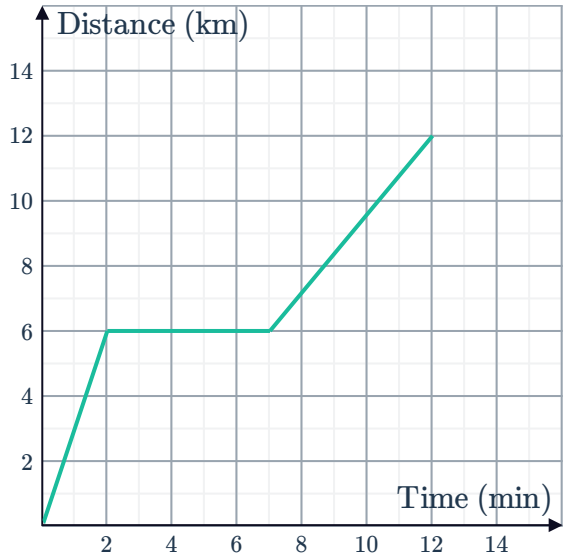
e How long, from the time she left home, did it take for Beth to return home?

f Find the total distance she travelled over the 9-minute period.



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10 The following travel graph shows the distance covered by Neil on his journey versus time. Neil took two trips and rested between them.

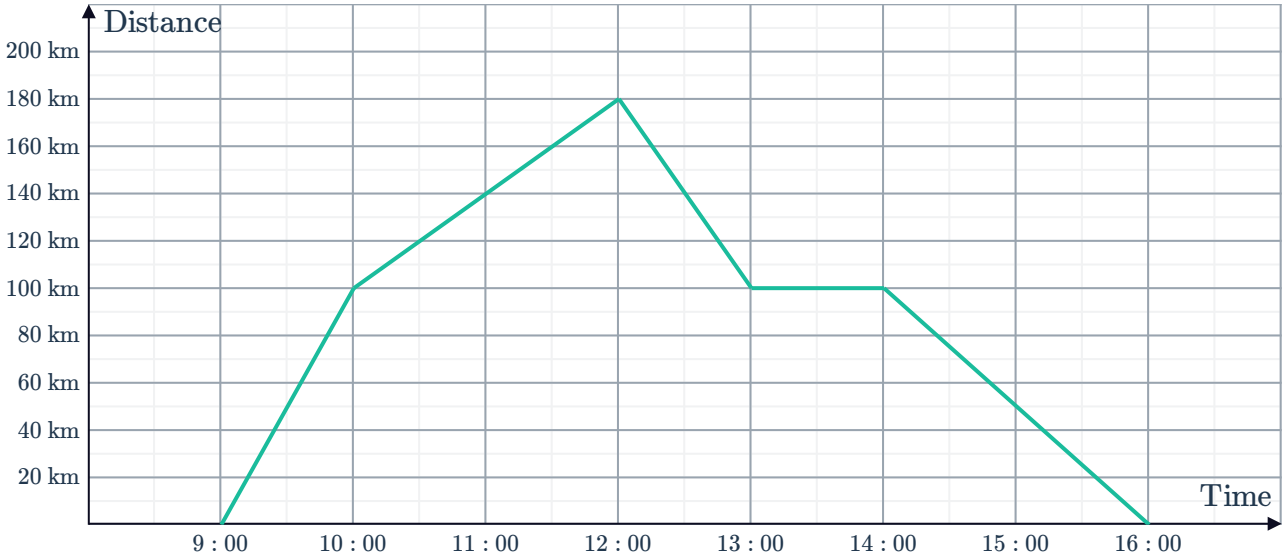


a Find the average speed of the first trip.

b Find the average speed of the second trip.

c Find the average speed of the entire journey.

11 The graph shows a train's distance from the central station throughout the day:



a When did the train change direction and start travelling back towards the central station?

b At what time did the train stop to change drivers?

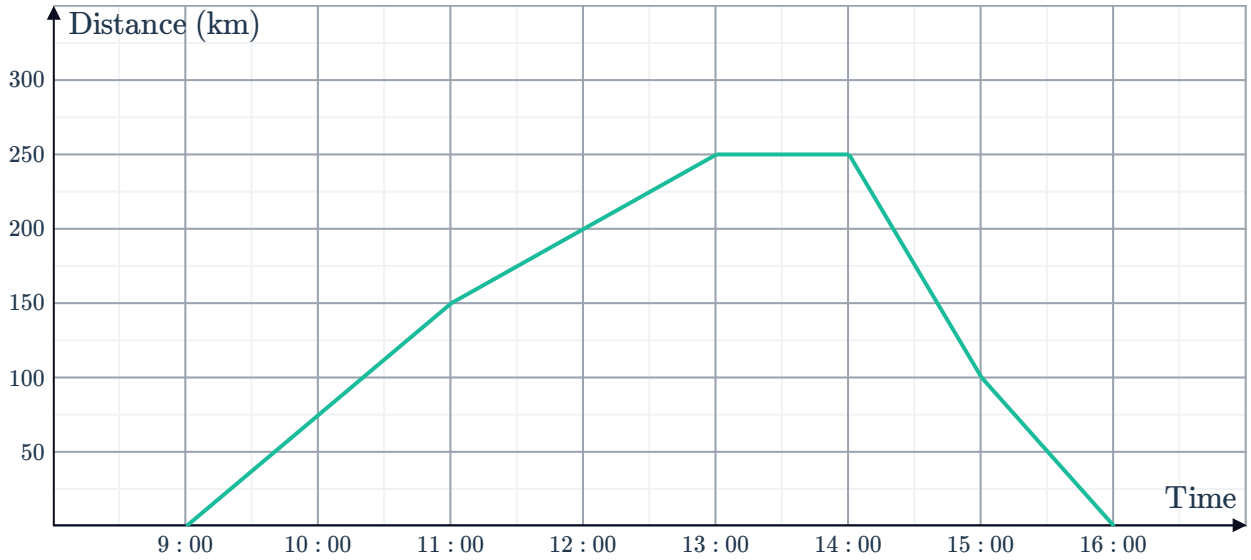
c How far had the train travelled between 12 pm and 1 pm?

d How far had the train travelled between 10 am and 4 pm?

e Between which times was the train was travelling fastest?

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12 The graph shows a traveller's distance from home each hour:

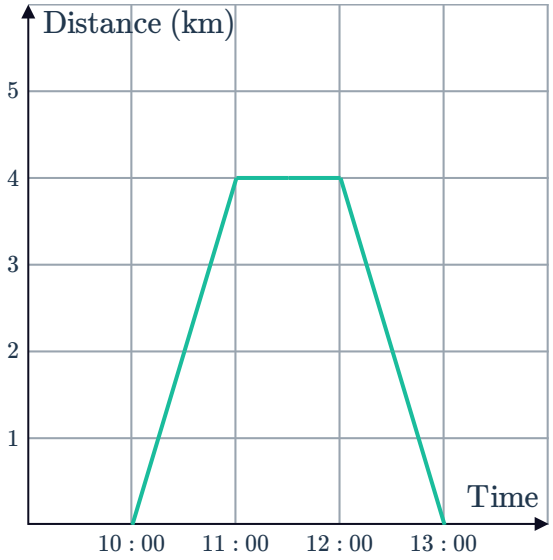


a At what speed was the person travelling between 11:00 am and 1:00 pm?

b In which interval of time was the person travelling fastest?

c What does the horizontal interval on the graph represent?

13 This line graph shows the distance Buzz was from his house as he travelled to visit his grandparents:



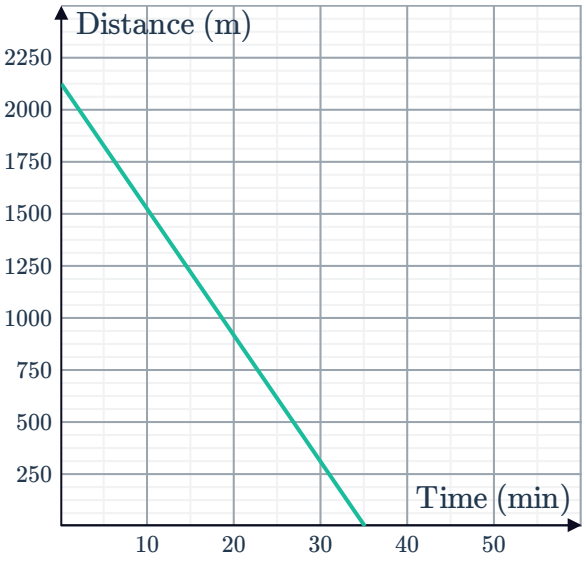
a How far was Buzz from his house at 10:30 am?

b What time did Buzz get back home?

c What was the furthest distance Buzz was from his house?

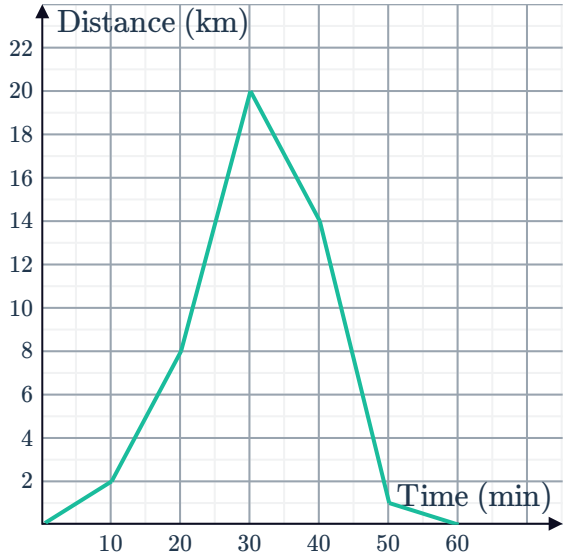
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14 Edward walks from his home to school on occasion. The graph represents his distance from school over one particular journey:



- **a** How far away is Edward's home from his school?
- **b** How long does Edward take to get to school?
- **c** How far is Edward from his school after 10 minutes?
- **d** How long did he have left in his trip when he was 1200 m away from his school?

15 During a test drive of an expensive car, the car's distance from the car dealership was tracked. The graph below shows how the distance changed over time:

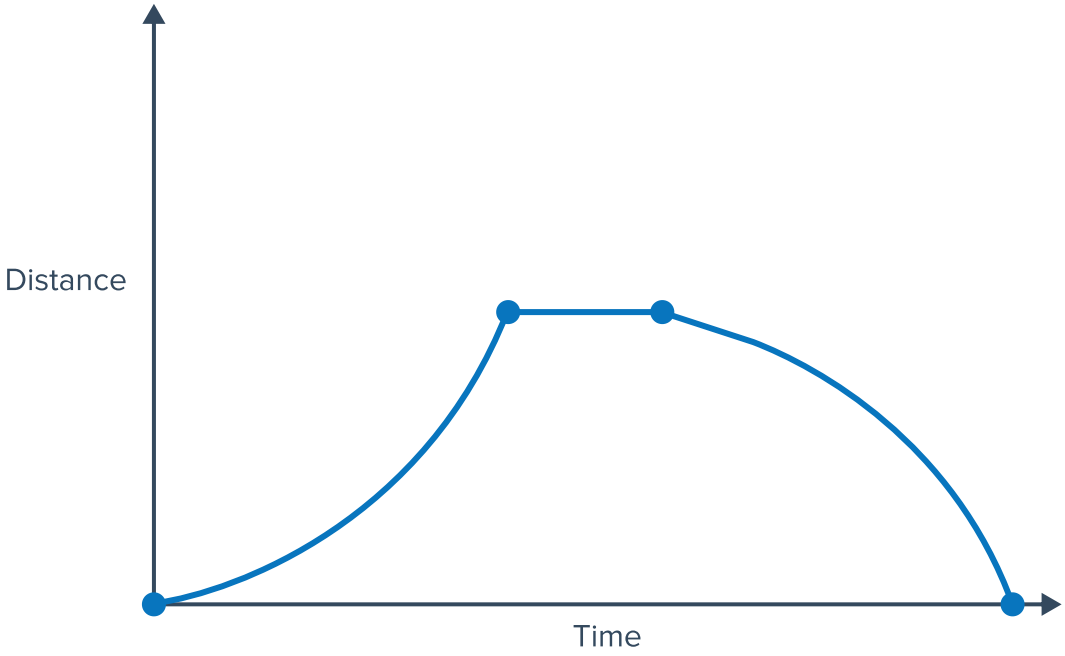


- **a** How long did the test drive take?
- **b** In which two 10-minute intervals was the driver travelling at the same speed?

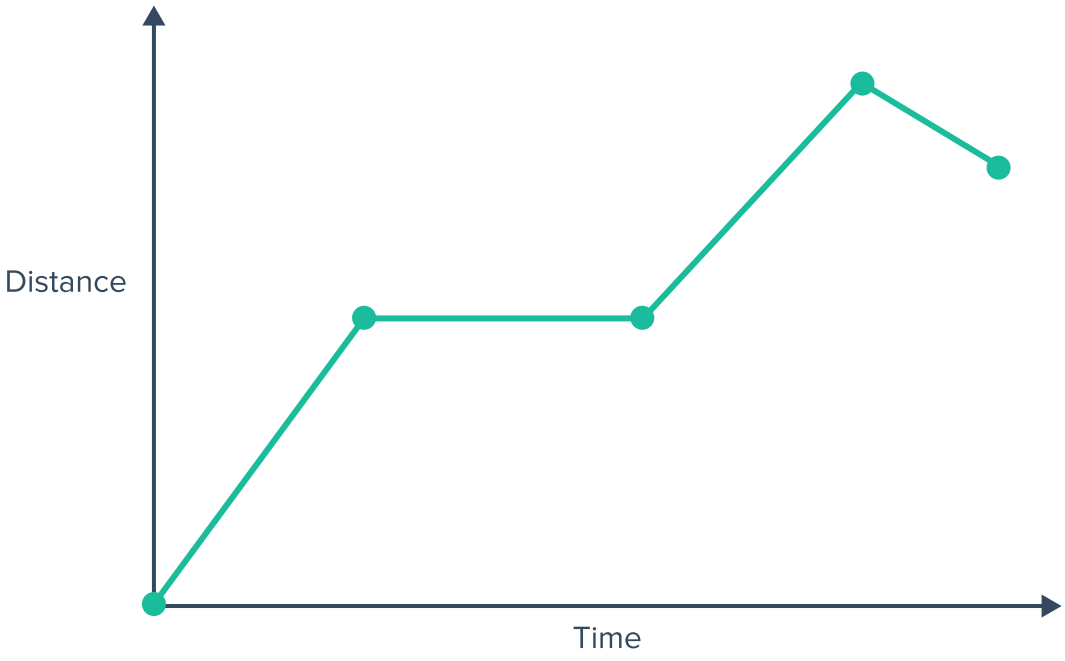
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16 Write a story to describe the following travel graphs:

a

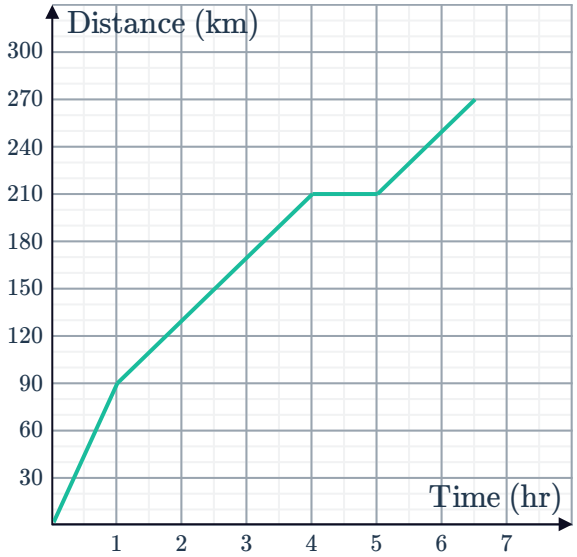


b



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17 Mr. and Mrs. Weber and their family travel $270\ \text{km}$ every year for their family trip. The following graph shows the travel distance and time:



a If the family leaves at 3 am, what time would they arrive at their destination?

b How far had they travelled after 5 hours?

c During the drive, the family stops for breakfast. At what time do they stop?

d At what times did their speed decrease?

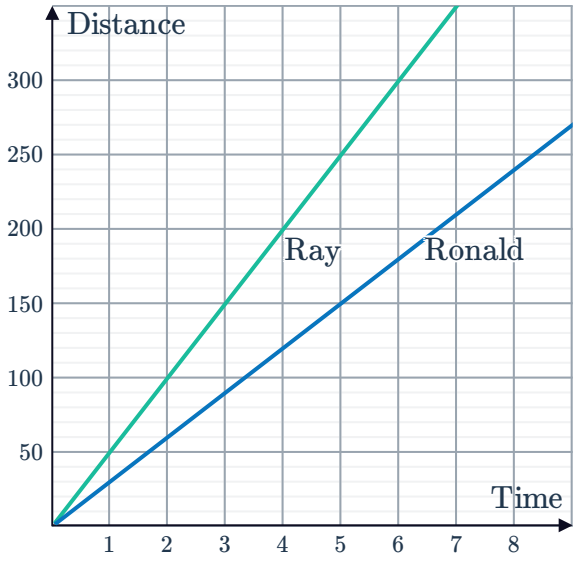
e The neighbours, the Axelrod family, have decided to join the Weber family on this trip but will drive there separately. For the Axelrod family's trip:

- They leave at the same time as the Weber family, but drive slower at first.
- They increase speed after 2 hours.
- They only stop for breakfast for half an hour.
- They arrive at the destination at the same time.

Sketch a graph showing both the Webers' trip and the Axelrods' trip.

Graph comparisons

18 The graph shows the progress of two competitors in a cycling race, where distance is in kilometres and time is in hours:

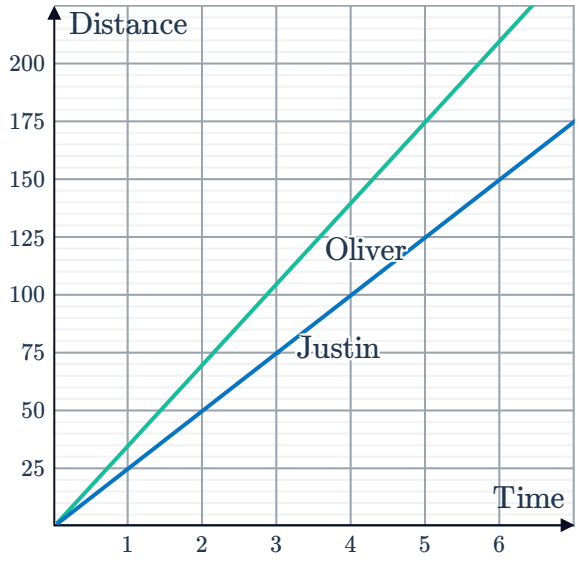


Who is travelling faster and by how much?

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19 The graph shows the progress of two competitors in a cycling race, where distance is in kilometres and time is in hours:

Who is travelling faster and how much faster is he travelling?



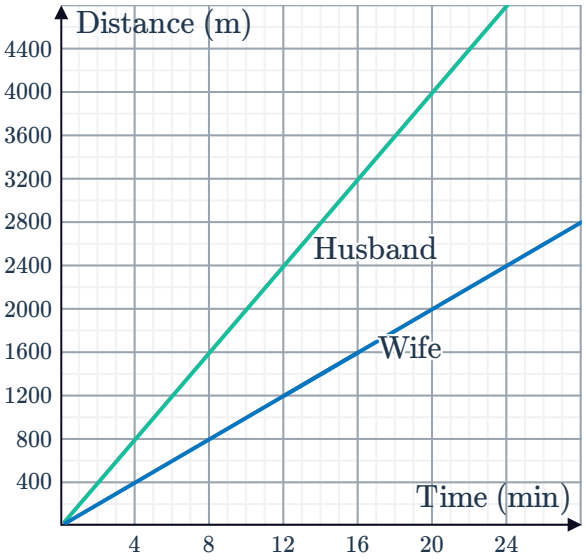
20 A husband and wife exercise each day for 20 minutes before dinner. The wife walks briskly, while the man runs. The distance each of them travel is shown on the graph:

a Find the difference in distance that each of them covers after 20 minutes.

b Find the distance the wife covers each minute.

c Find the distance the husband covers each minute.

d How long would it take the wife to walk the same distance that her husband runs in 6 minutes?



21 Two students were walking on a straight walking track and their travel graph is plotted:

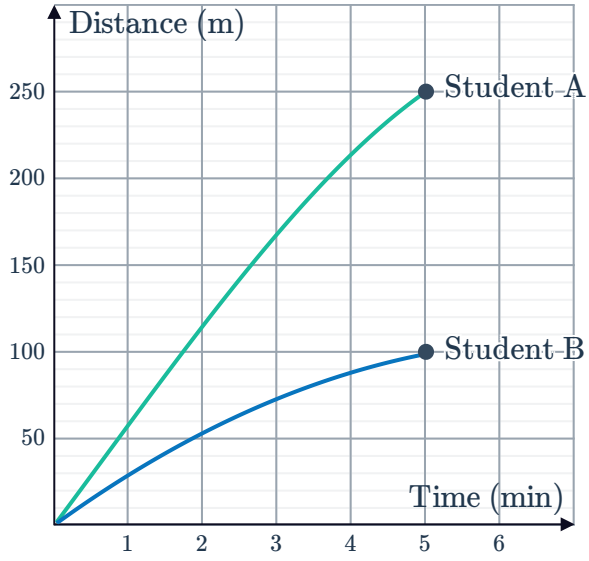
a How far did Student A walk?

b How many more metres did Student A walk than Student B during the first 5 minutes?

c What is the average speed of Student A in metres per minute?

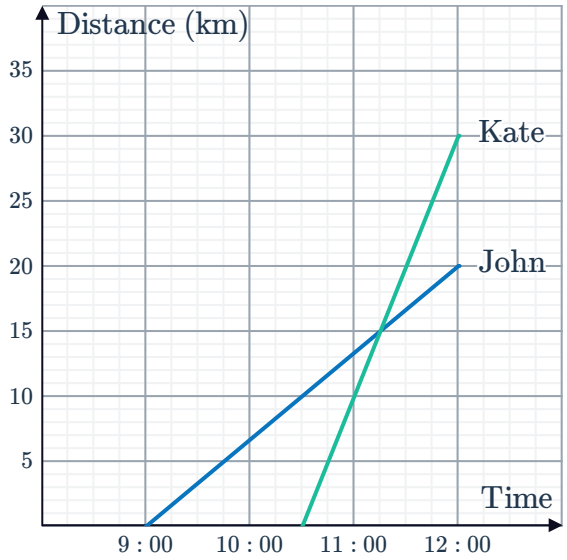
d What is the average speed of Student B in metres per minute?

e Assuming Student A and Student B walk at these speeds to complete the 2 km track on the field, how many minutes longer would Student B take than Student A?



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22 The travel graph of John and Kate is shown below:



a Calculate the difference in the amount of time travelled for John and Kate.

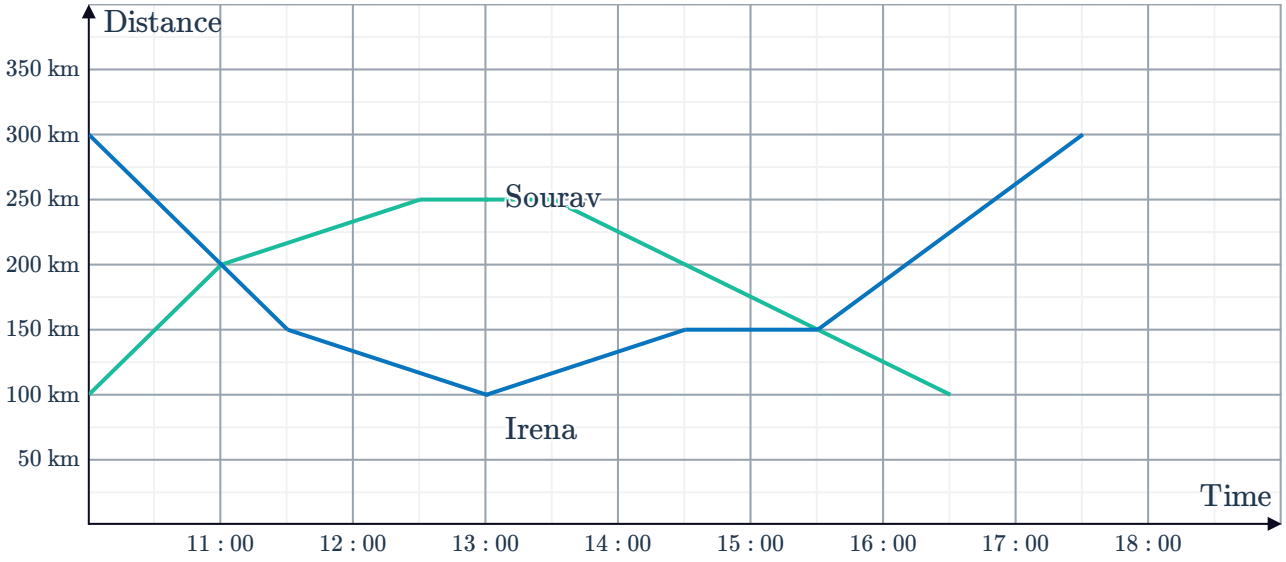
b Calculate John's average speed in km/h for the trip. Round your answer to two decimal places.

c Calculate Kate's average speed in km/h for the trip.

d At what time had Kate and John travelled the same distance?

e At the time when Kate and John had travelled the same distance, what was the actual distance travelled?

23 Sourav and Irena transport medical equipment from their respective work sites throughout the day. The graph shows their distance from home:



a If they are at their respective worksites at the beginning of the day, how far apart are their worksites?

b At what time are both Sourav and Irena the same distance away from their respective work sites?

c How far apart are they at 1 pm?

d At what time is Sourav $25\ \text{km}$ from arriving at his office?

e How long after Sourav returned to his worksite did Irena return to hers?

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